

Field Evaluation of Forced-Air Bubbling in a Single-Basin Solar Still under Karachi Climate: Productivity, Model Agreement and Life-Cycle Cost

Abdul Murad^{1*}, Muhammad Jamal¹, Sheikh Fahad¹, Abdul Hameed Memon¹

¹ Department of Mechanical Engineering, Faculty of Engineering Sciences and Technology, Hamdard University, Main Campus, Karachi, Pakistan

*Corresponding author: Abdul Murad | Email: muradjehan21@gmail.com

Abstract

Passive solar stills are attractive for decentralized freshwater production because they require simple hardware, yet their relatively low yield limits wider use. This study examines whether forced air bubbling can improve a conventional single-basin, single-slope solar still without changing its basic thermal architecture. Two comparable units were fabricated and tested outdoors in Karachi, Pakistan, from March to June 2024: one incorporated a perforated air line connected to an aerator, while the reference unit operated without air injection. The basin measured 1.2 m × 0.6 m, the glass cover was inclined at 19°, and the operating water depth was maintained at 0.02 m. Ambient temperature, solar irradiance and distillate production were recorded during the daily test periods. Monthly mean production for the aerated unit was 2.343, 2.421, 2.725 and 2.675 L/day for March, April, May and June, compared with 2.086, 2.163, 2.442 and 2.388 L/day for the reference unit. These pairs correspond to relative production increases of 12.31%, 11.89%, 11.60% and 12.02%, respectively. A source-project energy-balance calculation predicted 2.40 L/day for a validation case, compared with an experimental value of 2.39 L/day. In the project life-cycle estimate, the unit-water cost was PKR 4.98/L with aeration and PKR 5.38/L without aeration. The results show that modest air injection can raise freshwater yield in a simple solar still under Karachi conditions, although future work should quantify aerator power demand, measurement uncertainty and repeatability before scale-up.

Keywords: solar desalination; single-basin solar still; air bubbling; aeration; freshwater productivity

1. Introduction

Pressure on freshwater resources continues to increase as population, urban activity and industrial demand grow, while only a small proportion of the global water inventory is directly available for human use [1]. Desalination can supplement conventional supplies, but large-scale membrane and thermal processes may involve substantial energy use, infrastructure and operational expertise [2-4]. Solar-driven desalination is therefore of particular interest for small, decentralized applications where abundant solar energy can be used directly to support evaporation and condensation [5].

For basin-type solar stills, the central engineering problem is not process complexity but limited distillate output. Recent work has sought to improve that output through phase-change materials, modified absorbers, geometric optimization, photovoltaic-thermal integration and improved heat-management strategies [6-13]. These studies collectively show that solar-still productivity is strongly governed by the rates of heat transfer into the basin water and mass transfer away from the evaporating surface.

Air injection into saline water is not a new concept and should not be presented as one. Pandey reported in 1984 that forced air bubbling increased distillate output relative to a reference solar still [14]. Later work combined hot-air injection with other active modifications [15], investigated air-bubble humidification in desalination systems [16], and measured heat and mass transfer in a bubbled basin intended for solar desalination [17]. Because those studies used different configurations, operating temperatures, auxiliary components and performance metrics, their reported gains are not directly interchangeable with the present experiment.

The contribution of the present work is narrower and practical: a low-cost, side-by-side field comparison of two otherwise similar single-basin stills under Karachi weather, followed by a check against the source-project energy-balance calculation and a life-cycle cost comparison.

This framing avoids claiming novelty for the bubbling principle itself; instead, it evaluates how much benefit was obtained from a simple aeration retrofit in the specific experimental configuration available to the authors.

The objectives were therefore to (i) compare daily distillate production with and without forced air bubbling, (ii) examine whether the production advantage persisted across the March-June test period, (iii) compare one experimental yield with the theoretical source-model prediction, and (iv) quantify the project-estimated cost per litre for the two configurations.

2. Materials and Methods

2.1 Experimental solar-still configuration

Two single-slope basin stills were prepared for direct comparison. Each basin had a plan dimension of 1.2 m $\times$ 0.6 m, corresponding to an aspect ratio of 2. The front and rear wall heights were 0.06 m and 0.27 m, respectively, giving a mean water-to-cover spacing of 0.165 m. The glass cover thickness was 0.004 m and the cover inclination was 19°. The ceramic base thickness reported in the final-year project was 0.008 m. During testing, the water depth was maintained at 0.02 m. The modified unit contained a perforated air-delivery tube connected to an aerator; the reference still retained the same basic basin arrangement but operated without forced air injection.

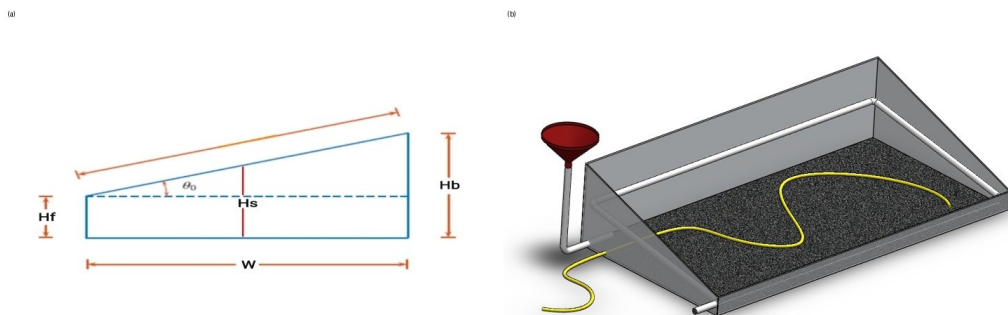


Fig. 1. Geometry and CAD representation of the single-basin solar still used in the source project.

Table 1. Principal design parameters reported for the experimental still.

Parameter	Value
Basin length	1.2 m
Basin width	0.6 m
Front wall height	0.06 m
Rear wall height	0.27 m
Mean specific height	0.165 m
Glass-cover slope	19°
Basin aspect ratio	2
Glass thickness	0.004 m
Ceramic base thickness	0.008 m
Operating water depth	0.02 m

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73 **2.2 Fabrication and test procedure**

74 The stills were assembled from the materials documented in the final-year project, including a
75 ceramic basin/base, a transparent glass cover, PVC piping and fittings, and silicone sealant. For
76 the modified unit, a perforated tube distributed air through the basin water. Before a test, saline
77 water was added to the basins, the units were positioned outdoors for solar exposure, and the
78 available measuring instruments were checked. The recorded test variables used in this paper are
79 ambient temperature, incident solar irradiance and collected distillate volume. The archival
80 project report does not provide a complete instrument-accuracy and uncertainty schedule;
81 consequently, no uncertainty bounds have been invented for the present manuscript.



Fig. 2. Outdoor experimental arrangement of the aerated and non-aerated solar stills during the test campaign.

2.3 Performance calculations and source-model comparison

Daily production was obtained from the accumulated distillate collected during the reported test period. For each month, the relative production increase due to aeration was calculated from the paired monthly mean values as $100(P_a - P_0)/P_0$, where P_a is the mean production of the aerated still and P_0 is that of the non-aerated still. The final-year project also contained component energy balances for the glass cover, basin water, basin liner and absorber, accounting for conductive, convective, radiative and evaporative terms. Rather than reconstructing missing symbols or constants, this paper reports only the source-project validation outcome that can be traced unambiguously: 2.40 L/day theoretical production versus 2.39 L/day measured production for the stated validation case.

2.4 Economic calculation

The life-cycle estimate in the source project assumed a 10-year service life and 290 operating days per year. Fixed costs were PKR 27,000 for the aerated unit and PKR 25,000 for the non-aerated unit. The project treated maintenance as 5% of fixed cost per year, which produced 10-year variable-cost totals of PKR 13,500 and PKR 12,500, respectively. Using 2.8 L/day for the aerated case and 2.4 L/day for the reference case, the reported life-cycle water costs were PKR 4.98/L and PKR 5.38/L. Electrical consumption of the aerator was not separately metered in the archived dataset, so the economic comparison should be read as the project-level estimate rather than a complete energy-cost assessment.

3. Results and Discussion

3.1 Hourly production behaviour

The paired hourly profiles show the same broad diurnal response in both units: distillate production increased from the morning toward the period of stronger solar input and then declined later in the day. The aerated unit remained above the reference unit through the main production interval in the representative 10-May-2024 record (Fig. 3). Because both units were exposed concurrently, the repeated separation between the two series is consistent with an aeration-related increase in evaporation rather than with a different daily solar cycle. The experiment, however, did not isolate bubble size, airflow rate or aerator power as independent variables, so the data support a system-level comparison rather than a mechanistic optimization of bubbling conditions.

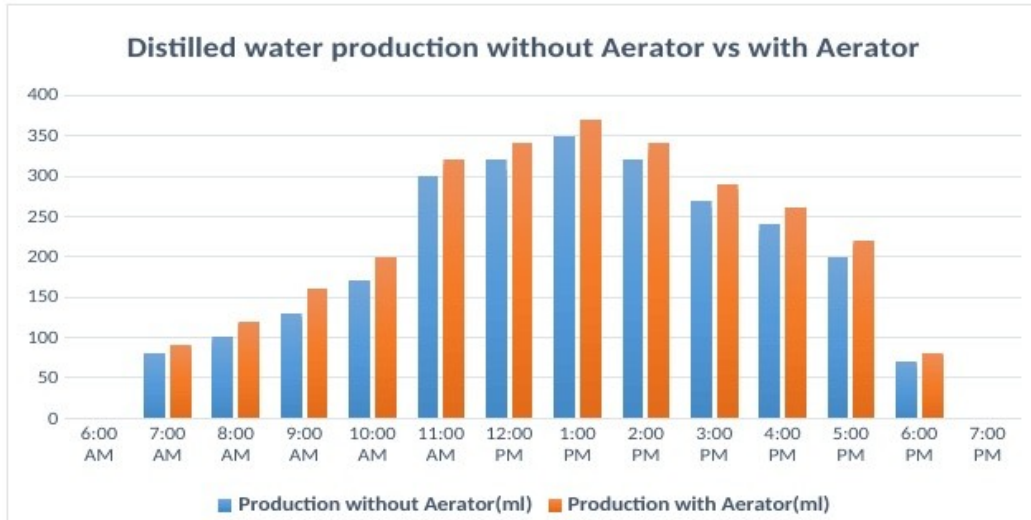


Fig. 3. Hourly distillate production with and without aeration on 10-May-2024, reproduced from the authors' experimental dataset.

3.2 Month-to-month performance

The monthly means demonstrate that the advantage of aeration was not restricted to a single test day. Mean production with aeration ranged from 2.343 to 2.725 L/day, while the corresponding reference values ranged from 2.086 to 2.442 L/day (Table 2). Calculated relative increases were 12.31% in March, 11.89% in April, 11.60% in May and 12.02% in June. May produced the largest absolute mean yield in both configurations. The source report associated the strong May performance with higher solar radiation and ambient temperature and noted that June coincided with increasing humidity and occasional cloudy conditions. These climatic explanations are plausible but should be treated qualitatively because the dataset does not include a multivariable statistical attribution analysis.

Table 2. Monthly mean distillate production and relative increase due to aeration.

Month	Without aeration (L/day)	With aeration (L/day)	Relative increase (%)
March	2.086	2.343	12.31
April	2.163	2.421	11.89
May	2.442	2.725	11.60
June	2.388	2.675	12.02

3.3 Relation to earlier bubbling studies

The observed 11.6-12.3% monthly production increase is smaller than some gains reported in earlier air-bubbling literature, but direct numerical comparison would be misleading. Pandey's early solar-still study reported a 33.5% increase for ambient-air bubbling and a larger increase when bubbling was combined with glass-cover cooling [14]. Eltawil and Omara used a substantially more complex active system that combined hot air with solar collectors, sprays or an external condenser, producing much larger gains under those operating modes [15]. Bubble-column humidification studies have also used elevated water temperatures, controlled airflow and dedicated humidifier geometries [16,17]. In contrast, the present project evaluates a comparatively simple retrofit applied to a basic field-operated basin still. Its result is therefore best interpreted as the measured benefit of that particular low-complexity configuration, not as a new bubbling mechanism or a universal enhancement factor.

3.4 Agreement with the source theoretical model

The archived validation calculation gave a daily theoretical yield of 2.40 L and a measured yield of 2.39 L for the same case, a difference of 0.01 L. This close numerical agreement indicates that the energy-balance formulation reproduced that test condition well. A stronger validation claim would require repeated comparisons across several days together with measurement uncertainty and model-error statistics. Because those elements are not available in the source record, the present paper deliberately limits the conclusion to agreement for the reported validation case.

3.5 Life-cycle cost comparison

The source economic calculation produced lifetime water costs of PKR 4.98/L for the aerated still and PKR 5.38/L for the non-aerated still. Although the aerated unit had the higher fixed cost, its assumed daily output was also higher, so the additional lifetime production reduced the calculated cost per litre. Figure 4 reproduces the project comparison. This result is useful as an initial feasibility indicator, but it does not include a separately measured electrical-energy term

for aeration and therefore should not be generalized to other aerator sizes, tariffs or duty cycles. More complete techno-economic work should meter blower or pump energy directly and report sensitivity to operating days, maintenance rate and component replacement.

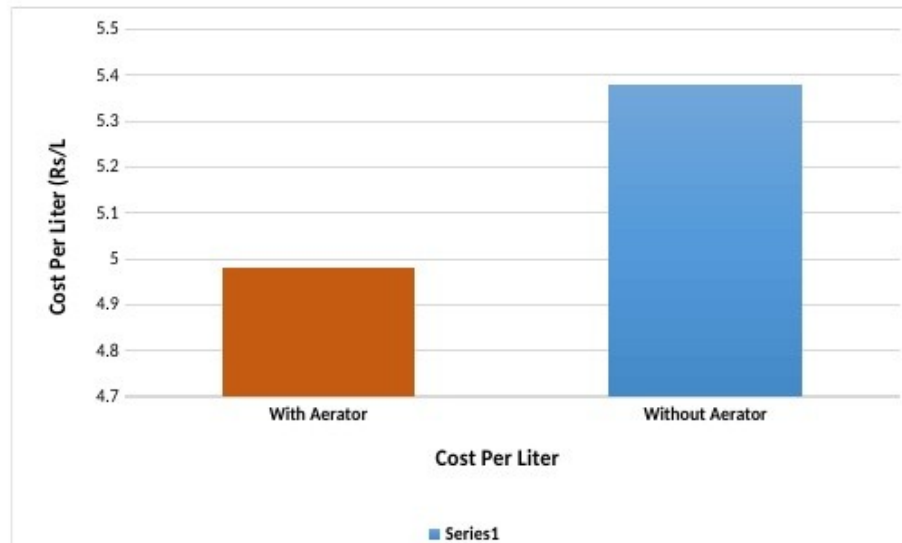


Fig. 4. Project-estimated cost per litre for the aerated and non-aerated solar stills.

3.6 Study limitations and data integrity

The field campaign provides a useful paired comparison, but several limitations should be carried into any interpretation or future submission. First, the archived FYP does not document replicate runs, uncertainty propagation or formal hypothesis testing. Second, airflow rate and aerator electrical power were not reported as controlled performance variables, so the experiment cannot identify an optimum aeration condition or a net energy benefit. Third, the work was performed at a single location over four months and does not establish annual performance. Finally, the FYP contains inconsistent post-treatment TDS values in different sections; for that reason, this manuscript does not use a single TDS value as a publication claim. Water-quality results should be rechecked against the original laboratory record before they are reported in a journal submission.

4. Conclusion

A forced-air bubbling retrofit was evaluated in a paired single-basin solar-still experiment conducted in Karachi from March to June 2024. Across the four monthly datasets, the aerated still produced 11.60-12.31% more distilled water than the non-aerated reference based on the reported monthly means. The project model also reproduced one daily validation result closely, giving 2.40 L/day compared with 2.39 L/day experimentally. In the archived life-cycle calculation, the higher assumed production of the aerated configuration reduced the estimated water cost from PKR 5.38/L to PKR 4.98/L despite its higher fixed cost. These results support air bubbling as a practical low-complexity enhancement for the tested basin-still configuration. They do not establish a new bubbling principle, nor do they yet quantify the energy penalty of aeration. A follow-on study should therefore measure airflow and aerator power, repeat tests with uncertainty reporting, verify product-water quality from the laboratory record, and evaluate seasonal performance before scale-up.

Acknowledgements

The authors acknowledge the supervision and laboratory support provided during the original final-year project at Hamdard University, Karachi.

Funding

No specific external research funding was reported in the source final-year project.

Data availability

The data summarized in this manuscript are taken from the authors' final-year project report and its associated experimental tables and figures. Availability of the underlying raw laboratory files should be confirmed by the corresponding author before journal submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this manuscript, the authors used OpenAI ChatGPT to improve language, organization, readability and journal-style presentation of material derived from the authors' final-year project. The tool was not used to create, alter or recalculate the experimental

measurements, figures or reported results. The authors reviewed and edited the resulting text and take full responsibility for the content of the manuscript.

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